CRISP-DM Plan for Insight Distiller: AI/ML Blog Summarizer

A comprehensive approach to developing an AI-powered tool that generates high-quality summaries of AI/ML blogs and articles, addressing information overload for professionals and e-learners in the rapidly growing field.

Introduction & Overview

Market Context

The global e-learning market is valued at $305B (2023) and projected to reach $1,516B by 2033 (Yahoo Finance, 2024), highlighting demand for efficient learning tools.

Target Audience

AI/ML practitioners, data scientists, and e-learners seeking timely, simplified insights from the overwhelming volume of content.

The rapid growth of AI/ML content (blogs, papers) overwhelms practitioners who lack time to read lengthy articles or rely on inconsistent manual summaries.

Problem Statement

> The rapid growth of AI/ML content (e.g., blogs, papers) overwhelms practitioners, who lack time to read lengthy articles or rely on inconsistent manual summaries.

Professionals need a tool that can efficiently distill key insights from technical content while maintaining accuracy and relevance, enabling them to stay current with trends and simplify complex information.

Objectives & Success Criteria

Business Objectives

- Deliver time-saving, relevant summaries to improve decision-making and learning
- Enable users to stay current with trends
- Simplify complex content
- Support tech companies in improving blog accessibility

Technical Success Criteria

Achieve higher ROUGE and BERTScore metrics compared to baseline DistilBART model:

Metric	Acceptable	Good	Excellent
ROUGE-1	0.30–0.35	0.36–0.45	> 0.45
ROUGE-2	0.08–0.12	0.13–0.20	> 0.20
BERTScore-F1	0.86-0.88	0.88-0.90	>0.90

Data Understanding & Preparation

Data Sources

- Blogs: Google AI Blog, Towards Data Science, Hugging Face Blog, Dev.to (ML tag)
- Abstracts: arXiv (AI category)

Data Preparation

- Cleaning: Remove HTML tags, emojis, special characters
- Normalization: Standardize text format
- Label Creation: Generate labeled summaries using OpenAI GPT-4o

Tools Used

- Collection: feedparser, requests, BeautifulSoup, Selenium
- Processing: pandas, nltk, transformers, tokenizers

Starting with 500 articles for MVP; scaling to multiple sources. Output: A clean dataset with columns: title, content, labeled_summary, stored in a pandas DataFrame.

Modeling Approach

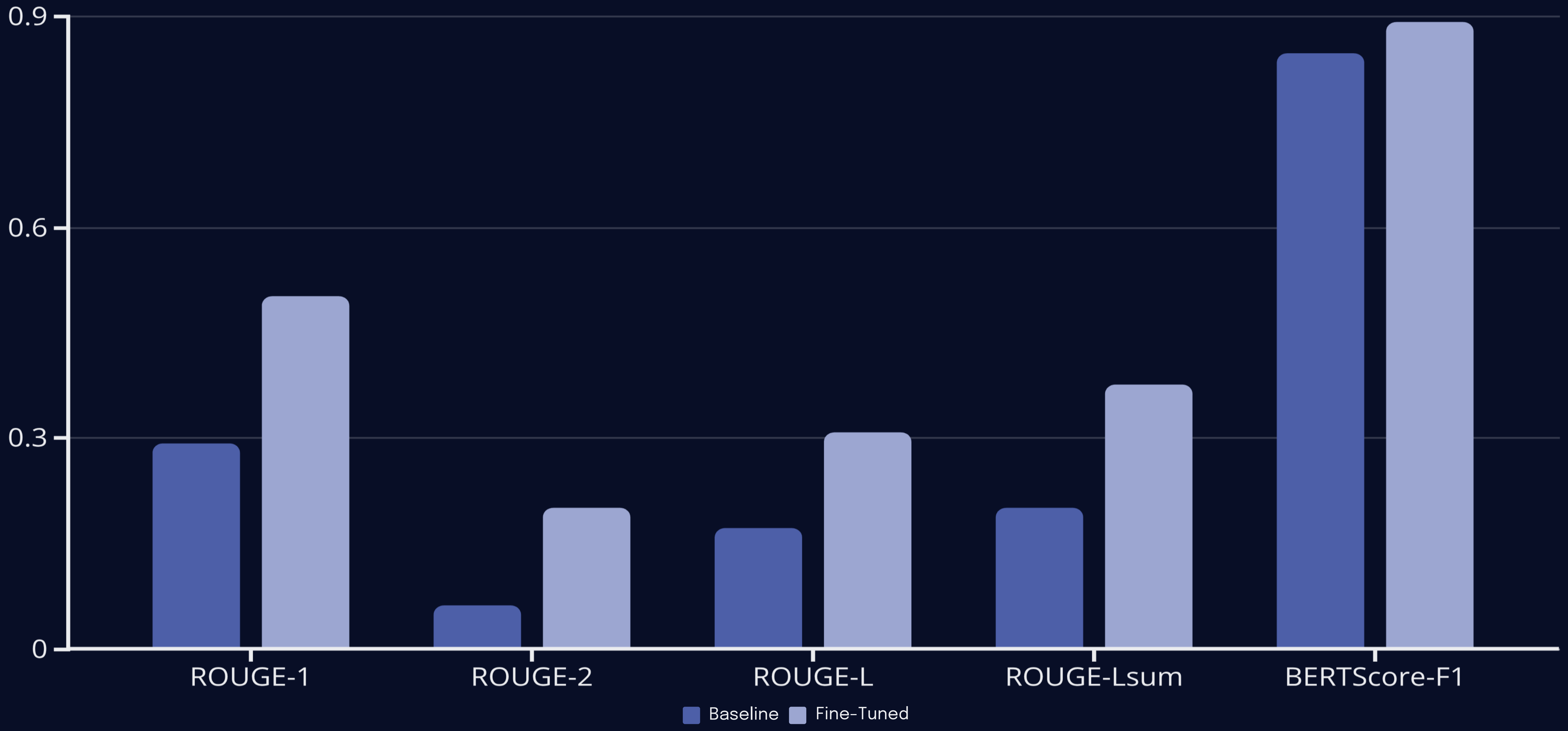
Base Model & Fine-Tuning

- Baseline: sshleifer/distilbart-cnn-12-6 (pre-trained on general text)
- Fine-Tuning: Transfer learning on AI/ML-specific dataset with GPT-4o labeled summaries
- Task Type: Sequence-to-sequence (text summarization, supervised learning)

Handling Long Inputs

- MAX_INPUT_TOKENS = 1024 (window size)
- CHUNK_OVERLAP = 50 (tokens reused between adjacent windows)
- Step size = 974 tokens
- Preserves local coherence with overlapping tokens

Findings & Analysis



The fine-tuned model shows significant improvements across all metrics, with ROUGE-1 and ROUGE-2 reaching "Excellent" thresholds, ROUGE-L reaching "Acceptable," ROUGE-Lsum reaching "Good," and BERTScore-F1 showing notable gains in the "Good" category.

Deployment Architecture

Internal FastAPI Service

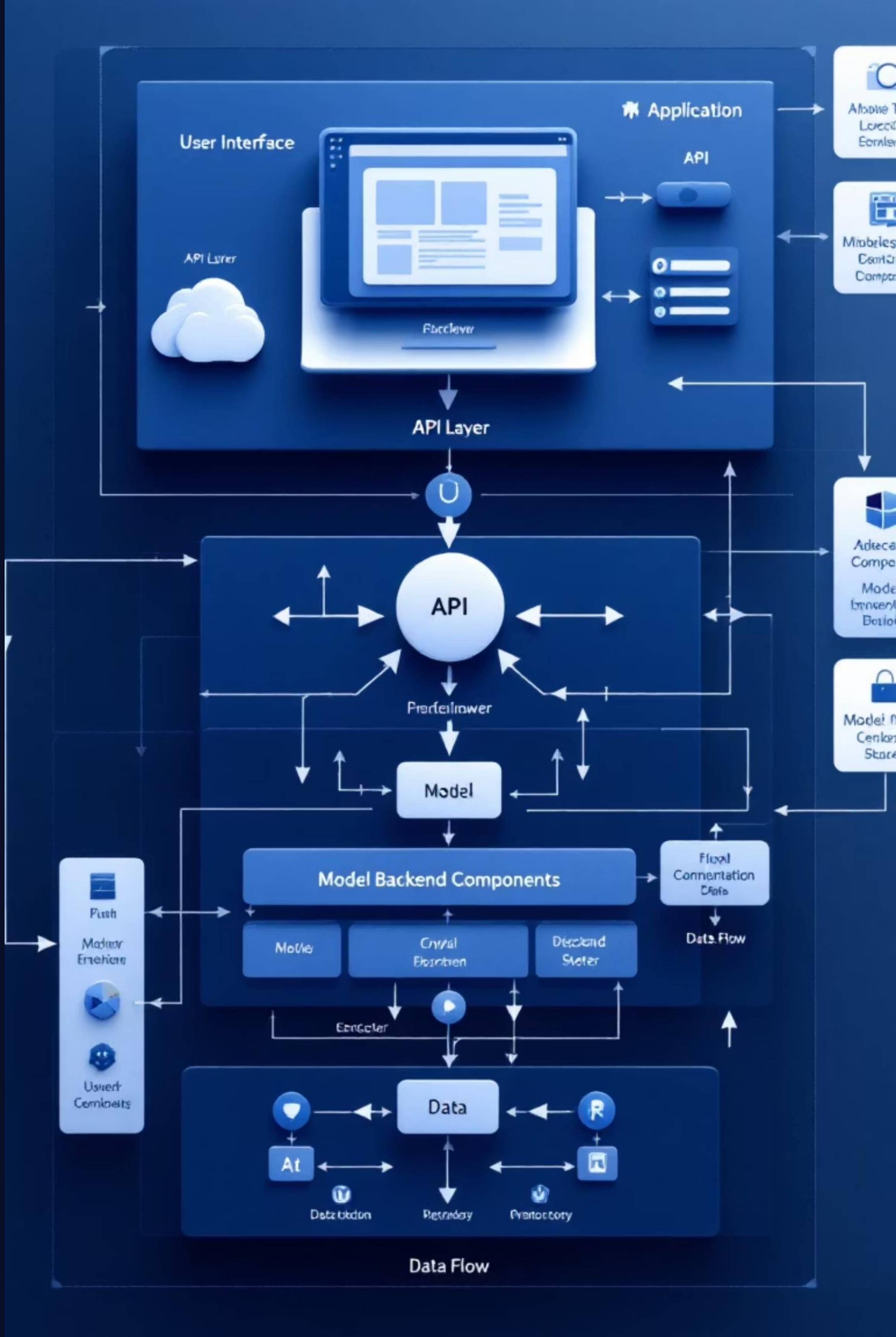
Loads the fine-tuned model, handles text processing, and runs inference with endpoints for text, PDF, and URL summarization.

Streamlit UI

Provides user interface with three tabs for different input types, sending requests to the internal API and displaying results.

Public API Gateway

Adds API-key security and provides stable, consumer-friendly routes for external applications.



Conclusion & Recommendations

Business Impact

For AI/ML practitioners, the improvements mean:

- Faster comprehension with higher trust
- More salient concepts captured
- Clearer phrasing and better organization
- Time savings when reviewing technical content

Recommendations

- Deploy the fine-tuned model behind the summarization API
- Gather user feedback to guide iterative updates
- Implement in-app ratings to validate quality beyond automated metrics
- Consider expanding to additional content types as the system proves successful

Future Work & Next Steps

1

Collect Human Ratings

Implement in-app feedback to benchmark perceived quality and drive active-learning loops for continuous improvement.

2

Explore Longer-Context Models

Investigate hierarchical chunking to reduce truncation effects and push ROUGE-L toward "Good" performance levels.

3

Optimize Generation Settings

Calibrate length penalties and beam settings per content type (blog vs abstract) for optimal readability.

4

Expand Training Data

Diversify with methods/experiments sections and different venues to increase model robustness.

Our Core Team

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This dedicated team brings a blend of expertise in AI, machine learning, and data science, ensuring the successful development and deployment of the Insight Distiller.